Modeling Norovirus Transmission on a College Campus Using the SIR-ious Epidemic Model

Feier Xu, Jiaying Zhang, Xiangyi Gao

MATH383L: Differential Equations Lab

April 27, 2026

Introduction

Norovirus, a highly contagious pathogen causing acute gastroenteritis, spreads rapidly in dense environments like university dormitories and dining spaces (Hall et al., 2013). Mathematical modeling provides a useful framework to study disease transmission and predict outbreak dynamics. One widely used approach is the SIR model, which classifies the population into susceptible (S), infected (I), and recovered (R) groups (Kermack & McKendrick, 1927). In this project, we apply the SIR model to simulate a norovirus outbreak on the UNC college campus using numerical methods implemented in Python.

Mathematical Model

The SIR model divides the total population $N = 1000$ into susceptible (S), infected (I), and recovered (R) compartments:

$$\frac{dS}{dt} = -\beta \frac{SI}{N}, \qquad \frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I, \qquad \frac{dR}{dt} = \gamma I$$

where β is the transmission rate and γ is the recovery rate. The system assumes a constant population $N = S(t) + I(t) + R(t)$ and captures the continuous transition of individuals: the susceptible pool is depleted by new infections ($-\beta SI/N$); the infected population grows simultaneously ($\beta SI/N$) but decays as individuals recover ($-\gamma I$); the recovered compartment increases at rate (γI).

Model Parameters and Initial Conditions

To simulate the outbreak within the campus community, model parameters and initial conditions were derived from clinical studies of nosocomial norovirus outbreaks, as detailed in Table 1.

Table 1: Definition and Justification of Baseline Model Parameters

Parameter	Definition	Value	Unit	Justification / Reference
N	Total Population	1000	Individuals	Estimated size of a typical residential college community.
$S(0)$	Initial Susceptible Population	990	Individuals	Assumes the vast majority of the population is initially susceptible.
$I(0)$	Initial Infected Population	10	Individuals	Represents the introduction of the virus via a small initial cluster.
$R(0)$	Initial Recovered Population	0	Individuals	Assumes no pre-existing immunity in the target cohort.
γ	Recovery Rate	0.5	Days^{-1}	Based on an average disease duration of 2 days (Vanderpas et al., 2009; Lynn et al., 2004).
R_0	Basic Reproduction Number	3.74	N/A	Estimated transmission potential in a closed, high-contact environment (Vanderpas et al., 2009).
β	Transmission Rate	1.87	Days^{-1}	Derived analytically from $\beta = R_0 \times \gamma$

The SIR system presented above uses coupled nonlinear equations. In order to be numerically stable and to capture the rapid acceleration towards the peak of the infection, we test two distinct numerical approaches.

Explicit Euler Method

The Explicit Euler method serves as our first-order numerical baseline. We discretize the time domain into *n* steps with a constant step size h, such that $t_{n+1} = t_n + h$. The state vector $y = [S, I, R]^T$ is updated using the first-order approximation: $y_{n+1} = y_n + h \cdot f(t_n, y_n)$, where f represents the system of ODEs:

$$S_{n+1} = S_n - h\beta \frac{S_n I_n}{N}, \quad I_{n+1} = I_n + h\left(\beta \frac{S_n I_n}{N} - \gamma I_n\right), \quad R_{n+1} = R_n + h\gamma I_n$$

As a simple method for approximating the actual solution, Euler's method assumes that the slope remains constant throughout the entire step h. However, failing to account for the curvature of the actual solution, it can be highly inaccurate and result

in a global truncation error of $O(h)$, which means the numerical error scales linearly with the step size. Given the highly contagious nature of norovirus, this error accumulates rapidly at larger step intervals, leading to instability and overestimation of the infection peak.

Fourth-Order Runge-Kutta Method (RK4)

To achieve higher precision, we implement the RK4 algorithm. This method approximates the solution by taking a weighted average of four incrementally evaluated slopes within each interval h:

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

where the increments are defined mathematically as:

$$k_1 = f(t_n, y_n), \quad k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right), \quad k_4 = f(t_n + h, y_n + hk_3)$$

Here, k_1 is the slope at the beginning of the interval, k_2 and k_3 are two successive midpoint slope estimates, and k_4 evaluates the slope at the end. The RK4 method has a global truncation error of $O(h^4)$. Consequently, halving the step size reduces the error by a factor of 16 (2^4), making it significantly more stable and accurate for capturing the acceleration of the norovirus epidemic without requiring an extremely small step size.

Numerical Implementation and Pseudocode

We employed a custom Python script to simulate the nonlinear SIR ODE system. A for loop was used to advance the state vector over the 30-day simulation

domain through our custom Explicit Euler and Fourth-Order Runge-Kutta (RK4) functions. To evaluate the global truncation errors without using external solvers, we established a high-accuracy reference baseline using our own RK4 implementation with a very small step size ($h = 0.001$). The stability and convergence of both methods at larger time step sizes were then tested against this reference solution. All data analysis and plots were generated using the matplotlib library.

Results and Discussion

To initialize a campus outbreak baseline, we set $\beta = 1.87$ and $\gamma = 0.5$ with a step size of $h = 0.001$. As we can observe from Figure 1, both the Fourth-Order Runge-Kutta (RK4) and Explicit Euler methods can capture the epidemic dynamics at this time resolution. The infected (I) population rises to nearly 400 within the first 5 days and then gradually drops as the susceptible (S) population becomes smaller. By day 15, most of the population has moved into the recovered (R) category, which shows that both methods can accurately capture this fast-spreading disease when the step size is sufficiently small.

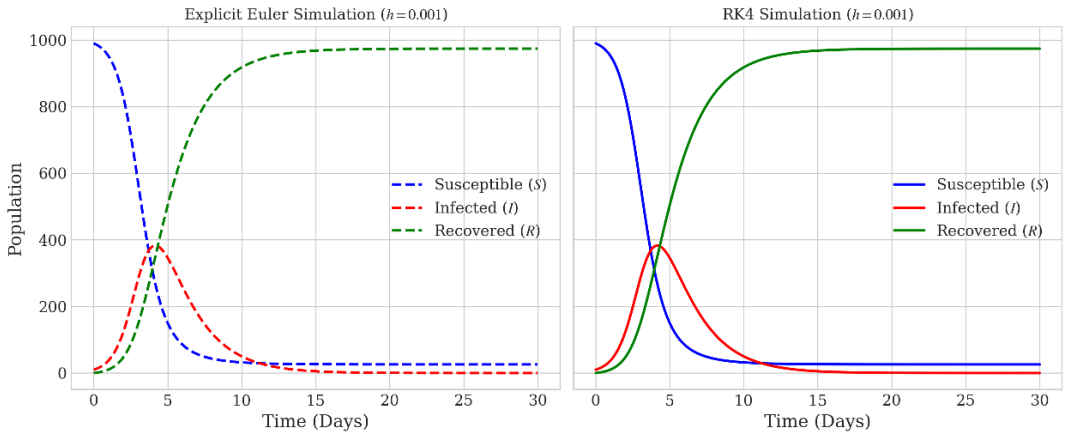


Figure 1. Baseline simulation of a campus norovirus outbreak ($\beta = 1.87, \gamma = 0.5$) with $h = 0.001$. Left: Explicit Euler method (dashed lines). Right: RK4 method (solid lines). Both yield nearly identical trajectories, providing a valid baseline.

Figure 2 displays the infected population (I) simulated with a small step size ($h = 0.001$) and a larger step size ($h = 1.0$) for each method. The Explicit Euler method becomes unstable with the larger step size. It overestimates the infection peak and shifts the timing of the outbreak because of accumulated truncation error. In contrast, RK4 remains stable at the larger step size, and its simulation stays close to the small-step baseline. Therefore, RK4 is a more reliable method for this high-transmission population, especially when larger step sizes are used.

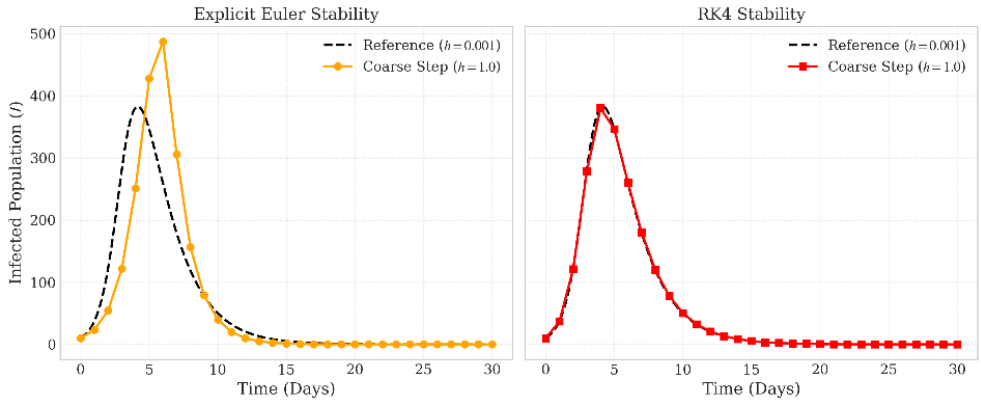


Figure 2. Stability comparison of the infected population (I) under fine ($h = 0.001$) and coarse ($h = 1.0$) step sizes. Left: The Explicit Euler method shows significant peak overestimation and numerical drift at $h = 1.0$. Right: The RK4 method remains stable, with its coarse-step trajectory staying close to the fine-step baseline.

In order to demonstrate the advantage of RK4 mathematically, we computed the global truncation error of the infected compartment $I(t)$ at $t = 10$, near the epidemic peak. With $h = 0.001$, we ran a high-accuracy RK4 simulation and used it as the reference solution for comparing both methods across different step sizes ($h = 0.1, 0.2, 0.25, 0.4, 0.5, 1.0$). As observed in the log-log plot, the Explicit Euler method shows first-order convergence, $O(h)$, whereas our Fourth-Order Runge-Kutta implementation follows fourth-order convergence, $O(h^4)$. This rapid decrease in error as the step size becomes smaller shows that RK4 is more accurate and more reliable

for solving this nonlinear SIR system than the Explicit Euler method.

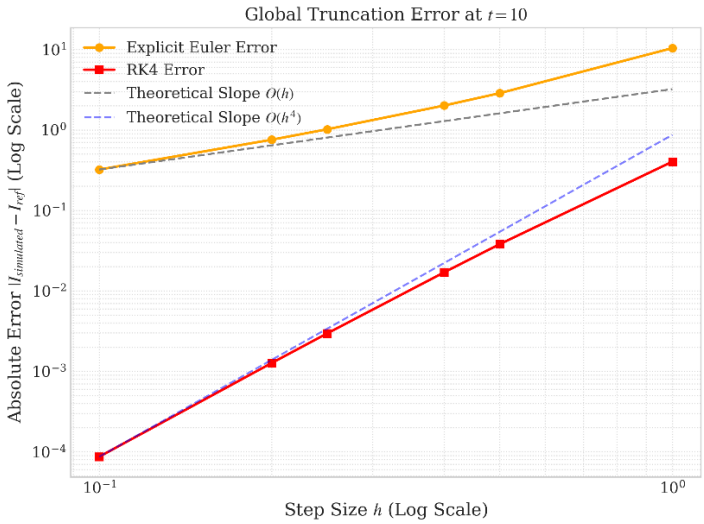


Figure 3. Log-log plot of the absolute global truncation error for the infected population $I(t)$ at $t = 10$. Using the high-precision RK4 simulation ($h = 0.001$) as the reference solution, the Explicit Euler method demonstrates first-order convergence $O(h)$, while the RK4 method is consistent with fourth-order convergence $O(h^4)$.

Conclusion

This study demonstrated how a norovirus outbreak in a high-density college environment can be modeled using the SIR epidemic model. By including real-world epidemiological parameters, $\beta = 1.87, \gamma = 0.5$, and $R_0 = 3.74$, we were able to capture the rapid spread of the pathogen. From a numerical perspective, the global error convergence results show the advantage of the Fourth-Order Runge-Kutta (RK4) method compared to the Explicit Euler method. While the first-order Euler scheme exhibits instability and overestimation in this nonlinear differential equation system, the $O(h^4)$ convergence of RK4 helps maintain accuracy and stability during the outbreak period. Future extensions could include heterogeneous dormitory networks or a quarantine compartment, making the model more useful for evaluating campus outbreak response strategies.

References

Hall, A. J., Lopman, B. A., Payne, D. C., Patel, M. M., Gastañaduy, P. A., Vinjé, J., & Parashar, U. D. (2013). Norovirus Disease in the United States. *Emerging Infectious Diseases*, 19(8), 1198–1205. https://doi.org/10.3201/eid1908.130465

Kermack, W. O., & McKendrick, A. G. (1927). A Contribution to the Mathematical Theory of Epidemics. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 115(772), 700–721. https://doi.org/10.1098/rspa.1927.0118

Lynn, S. J., Toop, J. M., Hanger, H. C., & Millar, N. D. (2004). Norovirus outbreaks in a hospital setting: The role of infection control. *The New Zealand Medical Journal*, 117(1189). http://www.nzma.org.nz/journal/117-1189/771/

Vanderpas, J., Louis, J., Reynders, M., Mascart, G., & Vandenberg, O. (2009). Mathematical model for the control of nosocomial norovirus. *Journal of Hospital Infection*, 71(3), 214–222. https://doi.org/10.1016/j.jhin.2008.11.024